

Chapter 21: Circuits & DC Instruments

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PHY201
Lecture 21



Part I

Resistors in Circuits

Outline of Part I: Resistors in Circuits

Series & Parallel Resistors

Outline of Part I: Resistors in Circuits

Series & Parallel Resistors

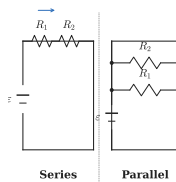
Two Ways to Connect Resistors

Series

- ▶ Connected **end-to-end**
- ▶ Same current through each
- ▶ Voltages add
- ▶ $R_s = R_1 + R_2 + \dots$

Parallel

- ▶ Connected **side-by-side**
- ▶ Same voltage across each
- ▶ Currents add
- ▶ $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

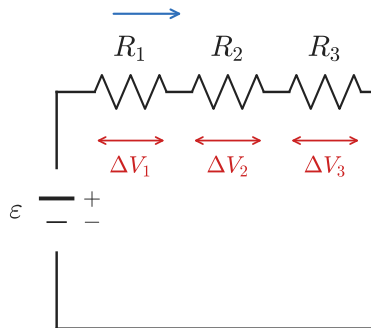


Series Resistors

Series Equivalent Resistance

$$R_s = R_1 + R_2 + \cdots + R_n$$

- ▶ Total resistance **increases** with each added resistor
- ▶ $R_s >$ any individual R_i
- ▶ Same current I flows through every element
- ▶ Voltages: $\Delta V_{\text{total}} = \Delta V_1 + \Delta V_2 + \cdots$



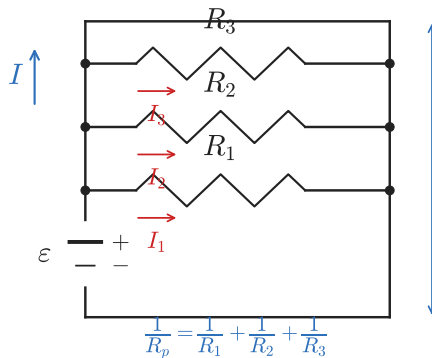
$$R_s = R_1 + R_2 + R_3$$

Parallel Resistors

Parallel Equivalent Resistance

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

- ▶ Total resistance **decreases** with each added resistor
- ▶ $R_p < \text{any individual } R_i$
- ▶ Same voltage ΔV across every element
- ▶ Currents: $I_{\text{total}} = I_1 + I_2 + \cdots$



Two Resistors in Parallel — Product-over-Sum Formula

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

This shortcut only works for **exactly two** resistors in parallel.

🔑 Adding a second path always *reduces* total resistance $\because$ current has more routes to take.

Two identical $6\ \Omega$ resistors in parallel:

$$R_p = \frac{6 \times 6}{6 + 6} = \frac{36}{12} = 3\ \Omega$$

Half the resistance of one resistor — makes sense!

Check Your Understanding

You add a third $10\ \Omega$ resistor to a circuit. Which configuration **increases** the total resistance?

- (A) Adding it in **series**
- (B) Adding it in **parallel**
- (C) Both increase resistance equally
- (D) Neither — total resistance stays the same

Come to lecture to see the answer.

Check Your Understanding

Three light bulbs are connected in **parallel** across a battery. Which quantity is the **same** for all three bulbs?

- (A) Current through each bulb
- (B) Voltage across each bulb
- (C) Power dissipated by each bulb
- (D) Resistance of each bulb

Come to lecture to see the answer.

Check Your Understanding

Three resistors ($5\ \Omega$, $10\ \Omega$, $15\ \Omega$) are connected in **series**. Which statement is true?

- (A) The $5\ \Omega$ resistor carries the most current
- (B) All three carry the same current
- (C) The $15\ \Omega$ resistor has the smallest voltage drop
- (D) Adding another resistor in series decreases total resistance

Come to lecture to see the answer.

Problem-Solving Strategy: Resistor Networks

Step-by-Step

1. Identify all series groups and parallel groups
2. Reduce parallel sub-networks first to get R_p
3. Combine R_p with any series resistors to get R_s
4. Find total current: $I = \Delta V / R_{\text{total}}$
5. Work backwards to find branch currents and voltages

Key Checks

- ▶ Voltage drops across series elements add to source voltage
- ▶ Branch currents in parallel add to total current

Example: Combination Circuit

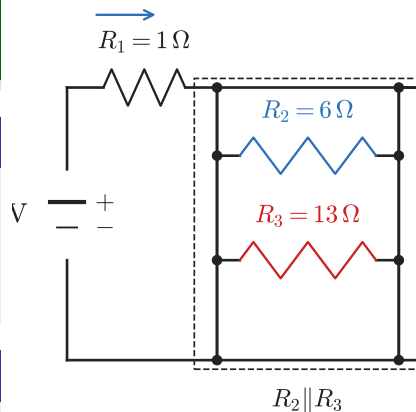
$R_1 = 1\ \Omega$ series, $R_2 = 6\ \Omega \parallel R_3 = 13\ \Omega$,
 $\Delta V = 12\text{ V}$

Step 1 — Parallel Equivalent

$$R_p = \frac{R_2 R_3}{R_2 + R_3} = \frac{(6)(13)}{6 + 13} \approx 4.1\ \Omega$$

Step 2 — Series Equivalent

$$R_s = R_1 + R_p = 1 + 4.1 = 5.1\ \Omega$$



Example: Combination Circuit (continued)

$$R_1 = 1\ \Omega, R_2 = 6\ \Omega \parallel R_3 = 13\ \Omega, \Delta V = 12\ \text{V}$$

Step 3 — Total Current

$$I = \frac{\Delta V}{R_s} = \frac{12}{5.1} \approx 2.35\ \text{A}$$

Step 4 — Voltages & Branch Currents

- ▶ $\Delta V_1 = IR_1 = (2.35)(1) \approx 2.35\ \text{V}$
- ▶ $\Delta V_p = 12 - 2.35 \approx 9.65\ \text{V}$ (same across R_2 and R_3)
- ▶ $I_2 = 9.65/6 \approx 1.61\ \text{A}; \quad I_3 = 9.65/13 \approx 0.74\ \text{A}$

Check Your Understanding

In the combination circuit (R_1 series, $R_2 \parallel R_3$), which resistor dissipates the **most power**?

- (A) R_1 — it carries all the current
- (B) R_2 — it has the smaller parallel resistance
- (C) R_3 — it is the largest resistor
- (D) Cannot be determined without more information

Come to lecture to see the answer.

Check Your Understanding

One branch of a parallel circuit is disconnected (open). What happens to the **total resistance** and the **total current**?

- (A) Resistance increases; current decreases
- (B) Resistance decreases; current increases
- (C) Resistance increases; current increases
- (D) Resistance decreases; current decreases

Come to lecture to see the answer.

Part II

EMF & Terminal Voltage

Outline of Part II: EMF & Terminal Voltage

EMF & Terminal Voltage

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EMF & Terminal Voltage

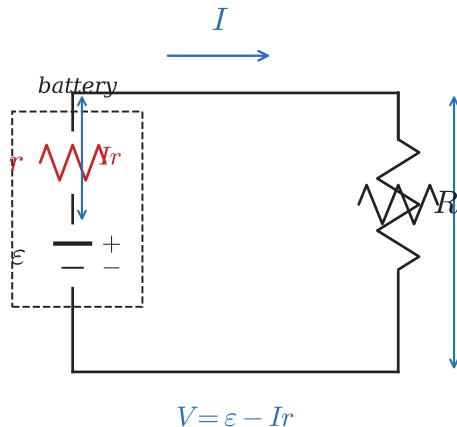
Electromotive Force (EMF)

- ▶ **EMF** (ε) is the *energy per unit charge* a source provides — not actually a force
- ▶ Sources: batteries, generators, solar cells, thermocouples
- ▶ Units: volts (V), same as potential difference

Ideal vs. Real Battery

An **ideal** battery delivers exactly ε to the circuit.

A **real** battery has *internal resistance* r that reduces the output.



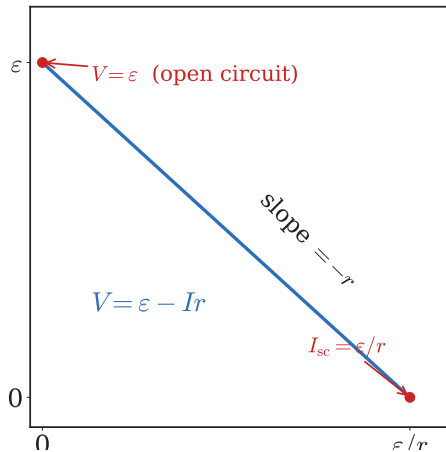
Terminal Voltage

Terminal Voltage

The voltage actually delivered to the external circuit:

$$V = \varepsilon - Ir$$

- ▶ ε = EMF of the source [V]
- ▶ I = current drawn [A]
- ▶ r = internal resistance [Ω]
- ▶ Ir = voltage “lost” inside the battery



Internal Resistance Effects

Small Load (large R_{ext})

- ▶ Small I
- ▶ $Ir \approx 0$
- ▶ $V \approx \varepsilon$
- ▶ Battery “looks” ideal

Heavy Load (small R_{ext})

- ▶ Large I
- ▶ Ir is significant
- ▶ $V \ll \varepsilon$
- ▶ Battery “droops”

Real-World Example

A car battery voltage drops when the starter motor (huge current draw) is engaged — interior lights dim momentarily.

Check Your Understanding

A battery has EMF $\varepsilon = 9\text{ V}$ and internal resistance $r = 1\ \Omega$. When **no current flows**, what is the terminal voltage?

- (A) Less than 9 V
- (B) Exactly 9 V
- (C) More than 9 V
- (D) Cannot be determined

Come to lecture to see the answer.

Check Your Understanding

As more devices are connected in **parallel** to a real battery, what happens to the terminal voltage?

- (A) It increases — more paths share the load
- (B) It stays the same — voltage is fixed by EMF
- (C) It decreases — more current increases the internal drop
- (D) It first increases, then decreases

Come to lecture to see the answer.

Check Your Understanding

Two identical 1.5 V batteries are connected in **series**. What is the combined EMF?

- (A) 0.75 V
- (B) 1.5 V
- (C) 3.0 V
- (D) 2.25 V

Come to lecture to see the answer.

Kirchhoff's Rules

Outline of Part III: Kirchhoff's Rules

Kirchhoff's Rules

Outline of Part III: Kirchhoff's Rules

Kirchhoff's Rules

When Do We Need Kirchhoff's Rules?

⚡ Simple series/parallel reduction **fails** for circuits with multiple sources or loops that cannot be collapsed into a single equivalent resistance.

Two Fundamental Laws for Any DC Circuit

- ▶ **Kirchhoff's Current Law** (KCL; Junction Rule) ← Conservation of Charge
- ▶ **Kirchhoff's Voltage Law** (KVL; Loop Rule) ← Conservation of Energy

Together, KCL and KVL provide enough equations to solve *any* DC circuit.

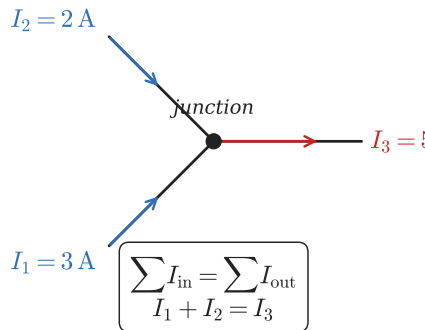
Kirchhoff's Current Law — Junction Rule

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

- ▶ Charge cannot accumulate at a junction
- ▶ Current entering = current leaving
- ▶ **Conservation of charge**

Example

$I_1 = 3 \text{ A}$ and $I_2 = 2 \text{ A}$ enter a junction.
 $\Rightarrow I_3 = 5 \text{ A}$ exits.



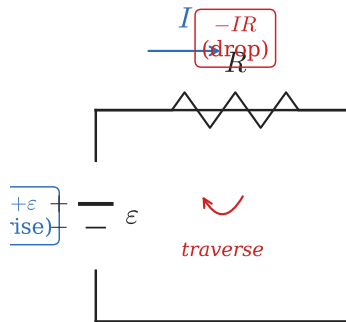
Kirchhoff's Voltage Law — Loop Rule

$$\sum \Delta V = 0$$

The sum of all voltage changes around any closed loop is zero.

Sign Conventions

- ▶ Resistor traversed *with* current: $-IR$
- ▶ Resistor traversed *against* current: $+IR$
- ▶ EMF source ($-$ to $+$): $+\varepsilon$
- ▶ EMF source ($+$ to $-$): $-\varepsilon$



$$\sum \Delta V = 0 \Rightarrow \varepsilon - IR = 0$$

Applying Kirchhoff's Rules — Strategy

Procedure

1. Label all branch currents with assumed directions
2. Apply junction rule at each independent node
3. Choose loops and traverse; apply loop rule
4. You need as many independent equations as unknowns
5. Solve the system; a negative result means the assumed direction was reversed

Tip

Choose current directions arbitrarily — the algebra corrects you if wrong.

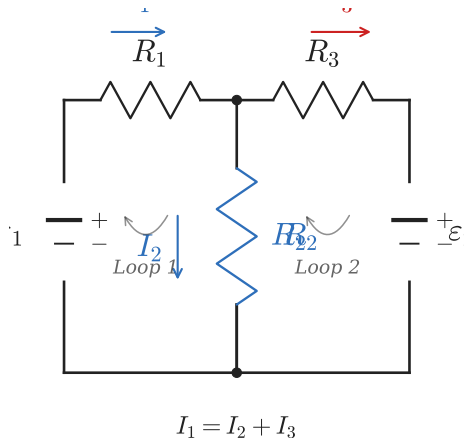
Example: Kirchhoff's Rules

Multi-Loop Circuit

Two EMF sources ε_1 , ε_2 with resistors R_1 , R_2 , R_3 in a two-loop network.

Setup

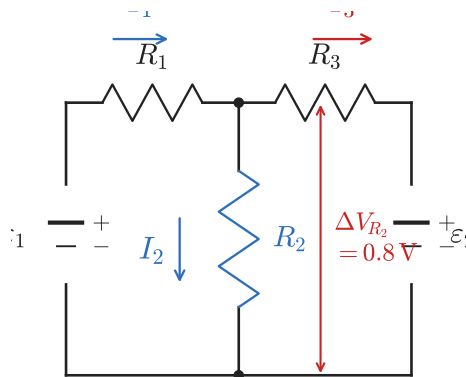
- ▶ Assign branch currents I_1 , I_2 , I_3
- ▶ Junction rule: $I_1 = I_2 + I_3$
- ▶ Loop 1: $\varepsilon_1 - I_1 R_1 - I_2 R_2 = 0$
- ▶ Loop 2: $\varepsilon_2 - I_3 R_3 + I_2 R_2 = 0$



Example: Kirchhoff's Rules — Solution

Key Takeaways

- ▶ Solve the system of equations simultaneously
- ▶ A negative current means the actual direction opposes your assumed direction
- ▶ Always verify: each loop sum should equal zero
- ▶ Junction currents must balance at every node



✓ Loops sum to zero ✓ Junctions balance

Kirchhoff's rules reduce to Ohm's law for

a simple single loop circuit

Check Your Understanding

At a circuit junction, currents $I_1 = 5\text{ A}$ and $I_2 = 2\text{ A}$ flow **in**, while I_3 flows **out**. What is I_3 ?

- (A) 3 A
- (B) 5 A
- (C) 7 A
- (D) 10 A

Come to lecture to see the answer.

Check Your Understanding

Traversing a loop you encounter a 12 V battery (from $-$ to $+$) and a $10\ \Omega$ resistor carrying I . Applying the loop rule: $12 - 10I = 0$. What is I ?

- (A) 0.83 A
- (B) 1.2 A
- (C) 120 A
- (D) 10 A

Come to lecture to see the answer.

Check Your Understanding

Kirchhoff's **junction rule** is a consequence of which conservation law?

- (A) Conservation of energy
- (B) Conservation of momentum
- (C) Conservation of charge
- (D) Conservation of mass

Come to lecture to see the answer.

Part IV

Measuring Circuits

Outline of Part IV: Measuring Circuits

DC Voltmeters & Ammeters

Null Measurements

Outline of Part IV: Measuring Circuits

DC Voltmeters & Ammeters

Null Measurements

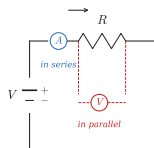
Measuring Voltage and Current

Voltmeter

- ▶ Measures **potential difference**
- ▶ Connected in **parallel**
- ▶ Must have very **high** R to draw negligible current

Ammeter

- ▶ Measures **current**
- ▶ Connected in **series**
- ▶ Must have very **low** R to cause negligible voltage drop



The Galvanometer

Core Meter Movement

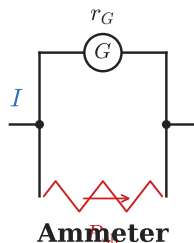
A **galvanometer** detects small currents via needle deflection in a magnetic field.

- ▶ Full-scale at current I_g (μA – mA range)
- ▶ Internal resistance r_G

Extending the Range

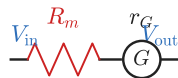
- ▶ **Ammeter**: add a small *shunt* resistor R_{sh} in *parallel* to bypass most current
- ▶ **Voltmeter**: add a large *multiplier* resistor R_m in *series* to drop most voltage

Ammeter



$$I_{\text{sh}} R_{\text{sh}} = I_g r_G$$

Voltmeter



$$R_m = \frac{V_{\text{range}}}{I_g} - r_G$$

Check Your Understanding

A voltmeter is accidentally placed **in series** with a resistor. What is the most likely result?

- (A) The voltmeter reads the correct voltage
- (B) The circuit current drops dramatically — nearly to zero
- (C) The resistor overheats
- (D) The voltmeter reads zero

Come to lecture to see the answer.

Check Your Understanding

An ammeter is accidentally placed **in parallel** with a resistor. What happens?

- (A) It accurately measures the current through the resistor
- (B) It creates a near short-circuit, potentially damaging the meter
- (C) Nothing — ammeters work in either configuration
- (D) The resistor burns out from too little current

Come to lecture to see the answer.

Check Your Understanding

An **ideal** voltmeter has _____ resistance, and an **ideal** ammeter has _____ resistance.

- (A) Zero; infinite
- (B) Infinite; zero
- (C) Equal; equal
- (D) High; high

Come to lecture to see the answer.

Outline of Part IV: Measuring Circuits

DC Voltmeters & Ammeters

Null Measurements

Null Measurement Methods

What Is a Null Measurement?

- ▶ Balance a known standard against an unknown quantity
- ▶ At the balance point (null): **no current flows through the detector**
- ▶ Meter draws no current $\Rightarrow$ measurement doesn't disturb the circuit
- ▶ More **precise** than direct measurement with a meter

Two Key Devices

- ▶ **Potentiometer** — measures unknown EMF
- ▶ **Wheatstone Bridge** — measures unknown resistance

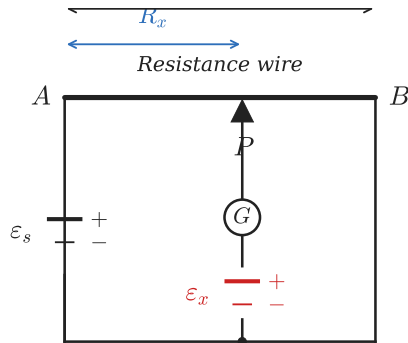
The Potentiometer

- ▶ Slide the contact until the galvanometer reads zero
- ▶ At balance, ratio of resistances = ratio of EMFs:

$$\varepsilon_x = \varepsilon_s \cdot \frac{R_x}{R_s}$$

Advantage

At null: zero current through the unknown source — no internal resistance error.



$$\text{At null: } \varepsilon_x = \varepsilon_s \cdot \frac{R_x}{R_s}$$

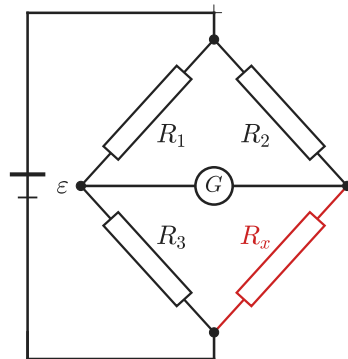
The Wheatstone Bridge

- ▶ Four resistors in a diamond arrangement; galvanometer across the middle
- ▶ Adjust known resistors until galvanometer reads zero (null)
- ▶ At balance:

$$R_x = \frac{R_2 R_3}{R_1}$$

Applications

Strain gauges, pressure sensors, temperature measurements — any transducer that changes resistance.



At null: $R_x = \frac{R_2 R_3}{R_1}$

Check Your Understanding

Why is a **null measurement** (e.g., the Wheatstone bridge) more accurate than a direct ohmmeter reading?

- (A) It uses more components, increasing precision
- (B) At balance, no current flows — the measurement doesn't disturb the circuit
- (C) It measures resistance directly using Ohm's law
- (D) It requires a digital display

Come to lecture to see the answer.

Part V

RC Circuits

Outline of Part V: RC Circuits

RC Circuits

Outline of Part V: RC Circuits

RC Circuits

Capacitors in DC Circuits

RC Circuit

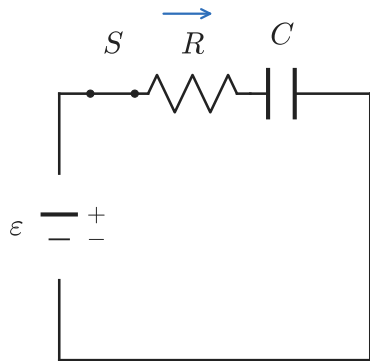
A circuit with resistor R and capacitor C in series with an EMF source.

- ▶ Capacitors store charge — don't respond instantly to voltage changes
- ▶ R controls how quickly charge flows onto/off C
- ▶ Behavior is **time-dependent** (exponential)

Two Phases

Charging: switch closed, capacitor fills up.

Discharging: source removed, capacitor empties through R .



$$\tau = RC$$

The Time Constant

Time Constant τ

$$\tau = RC$$

- ▶ Units: $[\Omega] \cdot [\text{F}] = \text{s}$
- ▶ τ sets the timescale for charging/discharging
- ▶ Larger R or larger $C \Rightarrow$ slower response

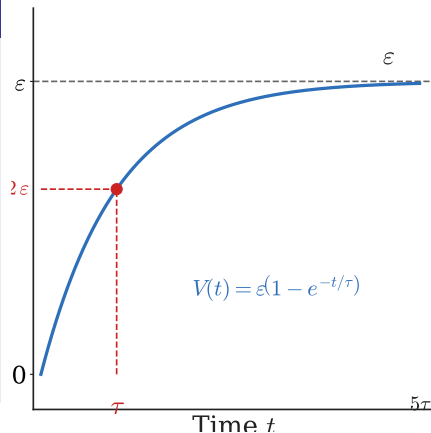
 After one time constant ($t = \tau$), the capacitor has reached $\approx 63\%$ of its final charge (charging) or dropped to $\approx 37\%$ of its initial charge (discharging).

Charging an RC Circuit

Voltage Across Capacitor While Charging

$$V(t) = \varepsilon(1 - e^{-t/\tau})$$

- ▶ Starts at $V(0) = 0$ (uncharged capacitor)
- ▶ Approaches $V(\infty) = \varepsilon$ (fully charged)
- ▶ At $t = \tau$: $V \approx 0.632 \varepsilon$
- ▶ After 5τ : $> 99\%$ fully charged

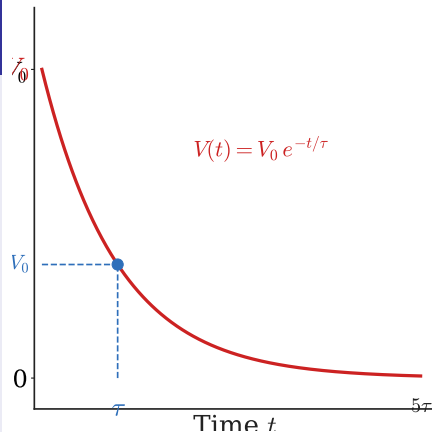


Discharging an RC Circuit

Voltage Across Capacitor While Discharging

$$V(t) = V_0 e^{-t/\tau}$$

- ▶ Starts at $V(0) = V_0$ (initial charge on capacitor)
- ▶ Decays toward zero exponentially
- ▶ At $t = \tau$: $V \approx 0.368 V_0$
- ▶ After 5τ : $< 1\%$ of initial charge remains



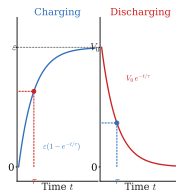
Charging vs. Discharging — Summary

Charging

- ▶ $V(t) = \varepsilon(1 - e^{-t/\tau})$
- ▶ $V: 0 \rightarrow \varepsilon$
- ▶ $I: I_0 \rightarrow 0$
- ▶ Capacitor acts like **short** at $t = 0$, **open** at $t \rightarrow \infty$

Discharging

- ▶ $V(t) = V_0 e^{-t/\tau}$
- ▶ $V: V_0 \rightarrow 0$
- ▶ $I: -I_0 \rightarrow 0$
- ▶ Capacitor acts as a **source**; current flows in reverse



Check Your Understanding

A capacitor charges through a resistor. After **one time constant** τ , the voltage across the capacitor is approximately what fraction of the EMF?

- (A) 37%
- (B) 50%
- (C) 63%
- (D) 99%

Come to lecture to see the answer.

Check Your Understanding

You double the resistance R in an RC charging circuit (keeping C and ε fixed). What happens to the **final equilibrium voltage** on the capacitor?

- (A) It doubles
- (B) It halves
- (C) It stays the same
- (D) It depends on the initial charge

Come to lecture to see the answer.

Check Your Understanding

At the **instant** a switch closes to charge an *initially uncharged* capacitor, it behaves like a _____. After a **very long time**, it behaves like a _____.

- (A) short circuit; open circuit
- (B) open circuit; short circuit
- (C) resistor; battery
- (D) battery; resistor

Come to lecture to see the answer.

Check Your Understanding

Which of the following is **not** a typical application of RC circuits?

- (A) Camera flash (slow charge, fast discharge)
- (B) Cardiac defibrillator timing pulses
- (C) Windshield wiper speed control
- (D) Measuring the resistance of a superconductor

Come to lecture to see the answer.

Chapter 21 Summary

Key Equations

- ▶ Series resistance: $R_s = R_1 + R_2 + \cdots + R_n$
- ▶ Parallel resistance: $1/R_p = 1/R_1 + 1/R_2 + \cdots + 1/R_n$
- ▶ Terminal voltage: $V = \varepsilon - Ir$
- ▶ Junction rule: $\sum I_{\text{in}} = \sum I_{\text{out}}$
- ▶ Loop rule: $\sum \Delta V = 0$
- ▶ Time constant: $\tau = RC$
- ▶ RC charging: $V(t) = \varepsilon (1 - e^{-t/\tau})$
- ▶ RC discharging: $V(t) = V_0 e^{-t/\tau}$

Chapter 21 Summary

Key Concepts

- ▶ Series R adds; parallel R gives $R_p <$ any individual branch
- ▶ Real batteries lose voltage under load: $V = \varepsilon - Ir$
- ▶ Kirchhoff's rules: charge conservation (junction) + energy conservation (loop)
- ▶ Voltmeter: parallel, high R ; Ammeter: series, low R
- ▶ Null measurements (potentiometer, Wheatstone bridge) eliminate meter loading error
- ▶ RC charging/discharging is exponential; $\tau = RC$ sets the timescale